

HTML

Day 1 – Introduction to HTML and Basic Structure

Topics to Cover:

- What is HTML?
HTML stands for **HyperText Markup Language**.
It is the standard language to create web pages.
- Basic Structure of an HTML Document:

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Webpage</title>
  </head>
  <body>
    <h1>Hello, World!</h1>
    <p>This is my first webpage.</p>
  </body>
</html>
```

Root Element

- **Definition:**
The **root element** is the **top-level element** that contains **all other HTML elements** in a page.
- In HTML, the **root element** is **<html>**.
- **Example:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Page</title>
  </head>
  <body>
    <h1>Hello World!</h1>
  </body>
</html>
```

Container Elements

- **Definition:**
Container elements are elements that **can hold other elements** inside them.
- They “contain” child elements.
- Most **block-level elements** act as containers.
- **Common container elements:**
 - <div> → generic container

- `<section>` → groups related content
- `<main>` → main content container
- `<header>`, `<footer>` → sections of a page

- **Example:**

```
<body>
  <div>
    <h1>Welcome</h1>
    <p>This is a container element example.</p>
  </div>
</body>
```

- Here, `<div>` is a **container element** containing `<h1>` and `<p>`.

3. Child Elements

- **Definition:**
Elements **inside a container element** are called **child elements**.
- They are “nested” inside a parent element.
- **Example:**

```
<div>
  <h1>Heading</h1>  <!-- child element -->
  <p>Paragraph</p>  <!-- child element -->
</div>
```

Visual Hierarchy

```
<html>      <- Root element
<body>      <- Container element
  <div>      <- Container element
    <h1>      <- Child element
    <p>      <- Child element
  </div>
</body>
</html>
```

Tip:

- Explain it like a **family tree**:
 - `<html>` = Grandparent (root)
 - `<body>` = Parent (container)
 - `<h1>`, `<p>` = Children (child elements)

{Don't write the tip in test paper, it just for understanding.}

- Key Parts:
 - `<!DOCTYPE html>` → declaration, Defines HTML version.

- `<html>` → Root element of the page.
- `<head>` → Metadata like title, links to CSS.
- `<body>` → Visible content on the webpage.

Practice Questions:

- What does HTML stand for?
- What are the two main sections of an HTML document?
- Write the basic structure of an HTML document with a title "My Site".
- What does the `<h1>` tag do?

Day 2 – Block-level vs Inline Elements, Common Tags

Topics to Cover:

- **Block-level Elements**
Take full width, start on a new line.
Example: `<div>`, `<p>`, `<h1>` - `<h6>`
- **Inline Elements**
Take only required width, stay in line.
Example: `<span>`, `<a>`, `<strong>`
- **Common Tags**
 - Paragraph → `<p>`
 - Headings → `<h1>` to `<h6>`
 - Division → `<div>`
 - Span → `<span>`

Example:

```
<h1>This is a heading</h1>
<p>This is a paragraph with <span>inline text</span>.</p>
<div>This is a block-level div element.</div>
```

Practice Questions:

- Name two block-level elements and two inline elements.
- How is `<div>` different from `<span>`?
- Write an example where you use both block-level and inline elements.

Day 3 – Semantic Tags & Formatting Tags

Topics to Cover:

- **Semantic Tags** → Give meaning to content:

- <header>, <footer>, <nav>, <section>, <article>, <aside>, <main>
- **Why Use Semantic Tags?**
→ Improves readability and accessibility.
- **Example:**

```
<header>
  <h1>Website Logo</h1>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">Contact</a>
</nav>

<main>
  <article>
    <h2>Article Title</h2>
    <p>Article content here.</p>
    <aside> tip : ...</aside>
    <aside> Did You Know?</aside>
  </article>
</main>

<footer>
  <p>© 2025 My Website</p>
</footer>
```

- **Formatting Tags:**
 - → Bold
 - <i> → Italic
 - <u> → Underline
 - → Important text (bold)
 - → Emphasized text (italic)
 - <mark> → Highlighted text
 - <small> → Small text

Practice Questions:

- What is the purpose of semantic tags?
- Write an example structure using <header>, <nav>, <main>, and <footer>.
- What is the difference between and ?
- How do you make text italic?

What are Interactive Tags?

- Interactive tags are HTML elements that allow **user interaction** such as clicking, typing, or submitting.
- They make web pages **dynamic and usable** (not just static text).

Common Interactive Tags

1. **<a>** – Anchor (Link)

Creates hyperlinks to other pages or sections.

```
<a href="https://example.com">Visit Example</a>
```

2. **<button>** – Button

Creates a clickable button.

```
<button>Click Me</button>
```

3. **<input>** – Input Field

Allows users to enter text, numbers, passwords, etc.

```
<input type="text" placeholder="Enter your name">
```

4. **<textarea>** – Text Area

Multi-line text input (for comments, feedback).

```
<textarea rows="3" cols="30" placeholder="Write your comment..."></textarea>
```

5. **<select>** and **<option>** – Dropdown

Creates a dropdown list.

```
<select>  
  <option>Option 1</option>  
  <option>Option 2</option>  
</select>
```

6. **<label>** – Label for Form Elements

Connects text to form inputs (better accessibility).

```
<label for="username">Name:</label>  
<input id="username" type="text">
```

7. **<form>** – Form

Groups input fields and buttons for submitting data.

```
<form>  
  <input type="email" placeholder="Enter email">  
  <button type="submit">Submit</button>  
</form>
```

Key Points

- Interactive tags = **elements that users can engage with**.
- Mostly used in **navigation (links, buttons)** and **forms (input, textarea, select, form)**.
- They are essential for making web pages **functional and user-friendly**.

Day 4 – Links, Images, Attributes, Forms

Attributes

HTML tags often have additional information called attributes. Attributes provide extra details about an element and are placed inside the opening tag.

Global Attributes

Attribute	Description
class	Specifies one or more class names for an element
id	Specifies a unique id for the element
style	Inline CSS style for the element
title	Text shown when hovering over the element
hidden	Hides the element from the page

Links

```
<a href="https://www.google.com">Go to Google</a>
```

Used with `<a>` tag

Attribute	Description
href	URL of the linked page
download	Indicates the resource should be downloaded instead of navigated

Images

```

```

Attribute	Description
src	Image URL
alt	Alternative text if image fails to load (important for accessibility)
width	Width of the image in pixels or percentage
height	Height of the image in pixels or percentage

Forms (Simple)

```
<form>
  <label for="name">Name:</label>
  <input type="text" id="name" placeholder="Enter your name" required>
  <button type="submit">Submit</button>
</form>
```

HTML Forms and Validation (Only HTML)

Key Points

- Use `<form>` to collect user input.
- HTML5 provides **built-in validation** (no JS needed).
- Validation attributes:
 - `required` → field must be filled
 - `type="email"`, `type="number"`, `type="url"` → checks format
 - `min`, `max`, `minlength`, `maxlength` → restrict input range/length
 - `pattern` → regex-based custom validation
 - `placeholder` → hint text inside input

```
<!DOCTYPE html>
<html>
<head>
  <title>Job Application Form</title>
</head>
<body>
  <h2>Job Application Form</h2>

  <form>
    <!-- Full Name -->
    <label for="fullname">Full Name:</label>
```



```
<input type="text" id="fullname" name="fullname" placeholder="Enter your full name"
      required minlength="3"><br><br>
```

```
<!-- Email -->
```

```
<label for="email">Email Address:</label>
```

```
<input type="email" id="email" name="email" placeholder="Enter your email"
required><br><br>
```

```
<!-- Phone -->
```

```
<label for="phone">Phone Number:</label>
```

```
<input type="tel" id="phone" name="phone" placeholder="123-456-7890"
      pattern="[0-9]{3}-[0-9]{3}-[0-9]{4}" required><br><br>
```

```
<!-- Position -->
```

```
<label for="position">Position Applied For:</label>
```

```
<select id="position" name="position" required>
```

```
  <option value="">--Select--</option>
```

```
  <option value="frontend">Frontend Developer</option>
```

```
  <option value="backend">Backend Developer</option>
```

```
  <option value="designer">UI/UX Designer</option>
```

```
</select><br><br>
```

```
<!-- Resume Upload -->
```

```
<label for="resume">Upload Resume:</label>
```

```
<input type="file" id="resume" name="resume" accept=".pdf,.doc,.docx" required><br><br>
```

```
<!-- Experience -->
```

```
<label for="exp">Years of Experience:</label>
```

```
<input type="number" id="exp" name="exp" min="0" max="40" required><br><br>
```

```
<!-- Cover Letter -->
```

```
<label for="cover">Cover Letter:</label><br>
```

```
<textarea id="cover" name="cover" rows="5" cols="40" placeholder="Write a short cover
letter..." required></textarea><br><br>
```

```
<!-- Buttons -->
```

```
<button type="submit">Submit Application</button>
```

```
<button type="reset">Clear</button>
```

```
</form>
```

```
</body>
```

</html>

This form uses only **HTML5 validation**:

- **Name**: must be 3–30 characters.
- **Email**: must be in valid email format.
- **Password**: at least 6 characters.
- **All required fields** must be filled.

Explanation:

- href links to other pages.
- src gives an image path.
- alt describes image for accessibility.
- placeholder shows a hint.
- required makes the field mandatory.

points:

- i. **Global attributes** are reusable on almost every tag.
- ii. Some attributes are **specific to tags** (like src, href, alt, action, etc.).
- iii. Attributes improve element behavior, appearance, and accessibility.

Practice Questions:

- Write an <a> tag linking to "<https://example.com>".
- What does the alt attribute do in the tag?
- Create a form with a text input and a submit button.
- How to make a form input field required?

Day 5 – Tables, Lists, Iframes & Embeds

Topics to Cover:

- **Tables**

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>John</td>
    <td>30</td>
  </tr>
</table>
```

Lists

- **Ordered List**

It presents a numbered list of items, which is useful when the order of the items is important.

```
<ol>
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ol>
```

- **Unordered List** →

It presents a bulleted list of items. This is used when the order of the items does not matter.

- **Iframes & Embeds**

```
<iframe src="https://www.example.com" width="300" height="200"></iframe>
```

```
<embed src="file.pdf" width="500" height="600">
```

Explanation:

- Tables display data in rows and columns.
- Lists display items in order or as bullets.
- Iframe embeds other webpages.
- Embed media like PDF.

Practice Questions:

- Write a table with 2 columns: Name and Age, and 2 rows of data.
- Create an unordered list with three favorite fruits.
- How do you embed a PDF file in HTML?
- What is the difference between <iframe> and <embed>?

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